

Christophe Hatterer

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SUMMARY

MSc candidate in Mechanical Engineering (Computational Science & Engineering) at ETH Zürich, completing an MIT thesis on mixed-variable Bayesian optimisation with Prior Fitted Networks for engineering design. Builds reproducible scientific-ML software in Python and PyTorch: data pipelines over large imaging datasets (X-ray CT) on HPC clusters, plus a maintainer-reviewed open-source contribution to IDEALLab/EngiOpt. Comfortable across the stack from physics-based simulation to GPU machine learning. Native French and English.

EDUCATION

Massachusetts Institute of Technology - MIT - DeCoDe Lab <i>MSc Thesis in Mechanical Engineering - Supervised by Prof. Faez Ahmed and Prof. Mark Fuge</i> Mixed-Variable Bayesian Opt. with PFNs for Engineering Design	Mar. 2026 – Exp. Sep. 2026 Boston, MA, USA
(Swiss) Federal Institute of Technology in Zurich - ETH Zürich <i>MSc in Mechanical Engineering, specialisation Computational Science and Engineering</i> Relevant coursework: Probabilistic Artificial Intelligence; Engineering Design Optimisation; Optimization and Machine Learning; High-Performance Computing for Engineers.	Sep. 2024 – Exp. Sep. 2026 Zürich, Switzerland
(Swiss) Federal Institute of Technology of Lausanne - EPF Lausanne <i>BSc in Mechanical Engineering</i> Specialising in Applied Mathematics	Sep. 2020 – Sep. 2023 Lausanne, Switzerland
Technical University of Delft - TU Delft <i>Student Exchange Year</i> Specialising in Aerospace and Aeronautical Engineering	Aug. 2022 – Jun. 2023 Delft, Netherlands
Island School <i>International Baccalaureate</i> Specialisation Maths, Physics, Chemistry, French	Jun. 2019 Hong Kong, Hong Kong SAR

PROFESSIONAL EXPERIENCE

Multi-Physics Simulations Engineer <i>Hitachi Energy Ltd. - Internship</i> - Implemented advanced physical models for self-healing properties of HV transformer capacitors using computational modelling techniques. - Developed multi-physics simulation frameworks to analyse electrical component reliability and optimise product performance parameters.	Aug. 2025 – Feb. 2026 Baden-Daetwill, Switzerland
Aerospace Engineer <i>Axalp Technologies - Internship</i> - Designed and optimised simulation models by analysing technical requirements and collaborating with cross-functional teams. - Developed dynamic simulations to enhance aircraft landing gear kinematics using computational modelling and system optimisation. - Led conceptualisation and development of a lightweight thermal cooling system for an Aircraft Auxiliary Battery Pack in Siemens NX.	Sep. 2023 – Jun. 2024 Olten, Switzerland
Project Assistant Director <i>Asia One Technics Ltd. - Internship</i> - Assisted Project Director and team to manufacture Solar Car Charging Canopy - Oversaw orders of structural part fabrications, with on-site travel for quality control. - Established specifications for plastic and aluminium part injection moulding/casting	Jun. 2019 – Sep. 2019 Hong Kong, Hong Kong

RESEARCH

Research Assistant <i>Computational Mechanics Group ETH Zürich</i> - Conducted research on X-ray computed tomography of mortar specimens, analyzing fracture propagation in heterogeneous materials. - Designed components for a torsional crack propagation experiment, enhancing reliability and flexibility. - Built a reproducible data-processing pipeline (open-source Python modules) for structural analysis of large X-ray imaging datasets, running on HPC cluster resources. - Supervision by Prof. Laura Lorenzis (ETH) and Dr. Pietro Carrara (ETH)	Sept. 2024 – Aug. 2025 Zürich, Switzerland
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- Extended generative models (SliceGAN, conditional GAN, multi-variable VAE) to the 3D domain for engineering geometries, implemented in PyTorch.
- Contributed work upstream to **IDEALLab/EngiOpt** open-source library as a merged pull request (~2,142 additions, 73 commits) reviewed by maintainers; conformed to repository ruff, mypy, and pre-commit CI.
- Supervision by Prof. Dr. Mark Fuge (ETH) and Dr. Florian Felten (ETH)

SKILLS & EXPERTISE

Programming	Software	ML / Technical	CAD	Languages	Sports
Python/PyTorch	Paraview	Machine Learning	Solidworks	French (Native)	Sail Racing
NumPy/SciPy	COMSOL	Bayesian Optimisation	Siemens NX	English (Native)	Skiing
C/C++ (OpenMP)	Ansys Workbench	Probabilistic ML / GPs	Team Center NX	Mandarin (Limited)	Hiking
Julia/Matlab	FEMAP	Foundation Models	Autocad	German (Limited)	Triathlon
PostgreSQL	Abaqus	High-Performance Computing	Catia V5		
Git, Docker, Linux	Weights & Biases	Reproducible ML / CI			

LEADERSHIP ROLES & EXPERIENCES

Mechanical System Advisor Jul. 2024 – Jan. 2025
aCentauri Solar Racing Zürich, Switzerland

- Advised 20+ team members across Aerodynamics, Mechanical, and Structures for BWSC 2025, a 3000km solar endurance race.
- Implemented GANTT-based project scheduling, budgeting, and systems engineering methodologies.
- Enhanced solar car Simcenter GFEM for precise load case simulations and optimised vehicle development.

Lead Chassis Engineer Oct. 2021 – Jun. 2022
NOVA Electric Racing Maassluis, Netherlands

- Led a team of 6 in prototyping an electric racing bike chassis for performance.
- Developed sustainable flax-fibre composite bodywork using DFM techniques.
- Conducted FEM analysis in Ansys Workbench for 20+ components, validating the motorcycle frame's structural integrity.

Finance Manager Oct. 2021 – Jan. 2023
PolySI - Integration Week Lausanne, Switzerland

- Managed budgets and logistics for events, ensuring smooth execution for incoming students.

PROJECTS

- **High-performance C++ N-body simulation** using Barnes-Hut algorithm
 - Reducing computational complexity from $O(n^2)$ to $O(n \log n)$ for large-scale applications
- **Python-based transient analysis of landing gear structural dynamics**
 - Incorporating numerical dissipation techniques to enhance simulation stability by 45%
- **Bayesian Neural Network with SWA-Gaussian for satellite image classification**
 - Reducing false positives while quantifying prediction uncertainty for critical decision-making
- **Optimised drug candidate features through Bayesian Optimisation**
 - Achieving predictions of bioavailability while maintaining synthesisability constraints.
- Developed **Gaussian Process model for environmental PM2.5 prediction** in residential areas
- Performed **Multi-Objective NSGA optimisation of battery enclosures** using parametric design
- Investigated **Topology optimisation** within two-scale damaged structures.
- Conducted **Spectral/Modal analysis** of aluminium plates through reconstructive testing, identifying critical resonance frequencies.

Volunteering

- Nepal Service Trip – Orphanage Support – Kathmandu, Nepal (2018).
- HK Asia Pacific Adventure – Sports Instructor – Hong Kong (2017).
- Vietnam Service Trip – Library Construction – Hanoi, Vietnam (2016).

References available upon request.